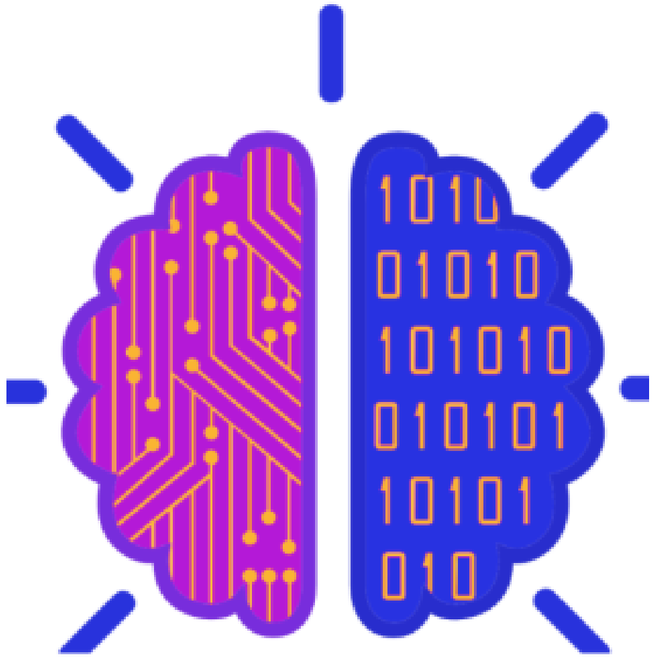
SCHEDULO - AI BASED TIMETABLE GENERATOR

Problem Statement Title: **AI-Based Timetable Generation System aligned with NEP 2020 for Multidisciplinary Education Structures**

Theme: **Smart Education**

PS Category: **Software**

Team Name: **CTRL + ALT+Z**

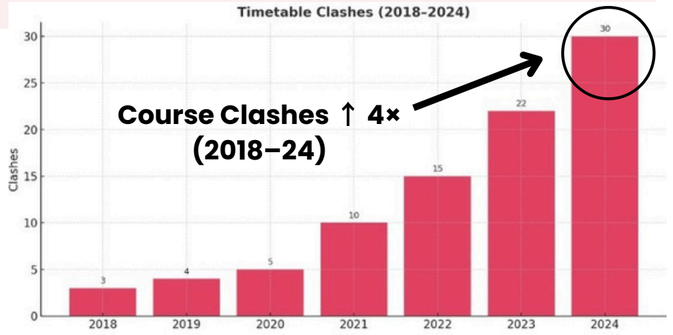


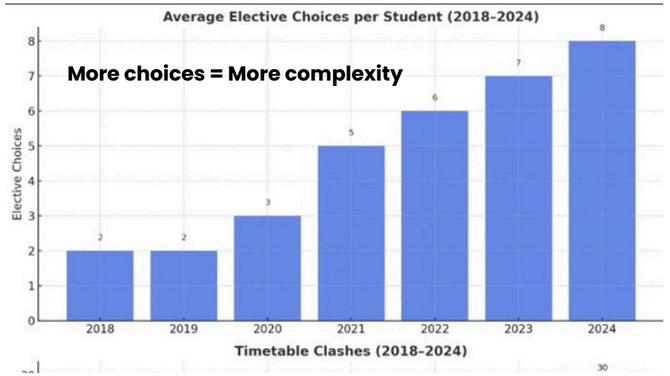
IDEA APPROACH

CHALLENGES

Manual timetable creation is slow, error prone and often leads to course conflicts. Existing systems fail to consider student preferences, attendance, or dynamic optimization for rescheduling.

- Elective bottlenecks: Popular courses oversubscribed, while unpopular courses are underused
- Policy pressure: NEP 2020 flexibility breaks rigid systems
- Equity gap: Late or limited access causes conflicts
- Cross-department clashes: Interdisciplinary electives overlap cores.

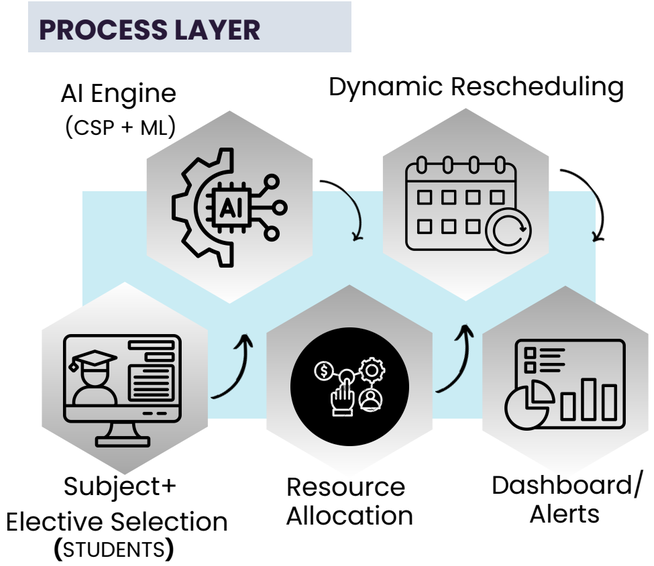




PROPOSED SOLUTION

SCHEDULO: AI-based Timetable Generator (Aligned with NEP 2020)

- **With NEP 2020's multidisciplinary framework, "SCHEDULO" auto-generates conflict-free timetables with an AI constraint solver and dynamic rescheduling.**
- **Student preference dataset + existing ERP APIs**



Unique Value Proposition

1. Integrated with **AI-Driven Experiential Scheduler** with Smart Attendance which Optimizes academic & experiential activities ensuring clashfree scheduling and real-time attendance tracking.

2. India's first NEP 2020 aligned timetable generator resolving elective clashes through **AI-driven dynamic rescheduling**

Unique Selling Proposition

Schedulo uniquely **balances faculty workload, integrates legacy ERPs,** and **adapts instantly** to policy or curriculum changes

TECHNICAL APPROACH

Schedulo

PROTOTYPE

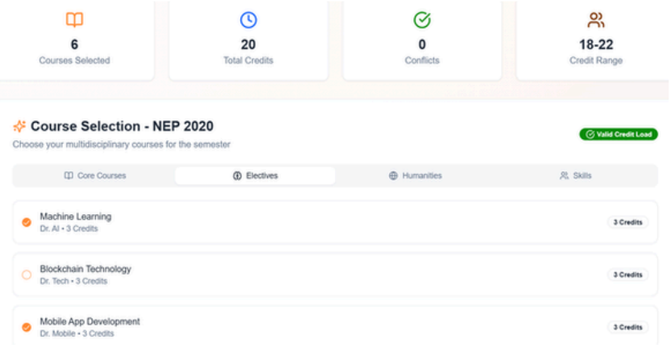
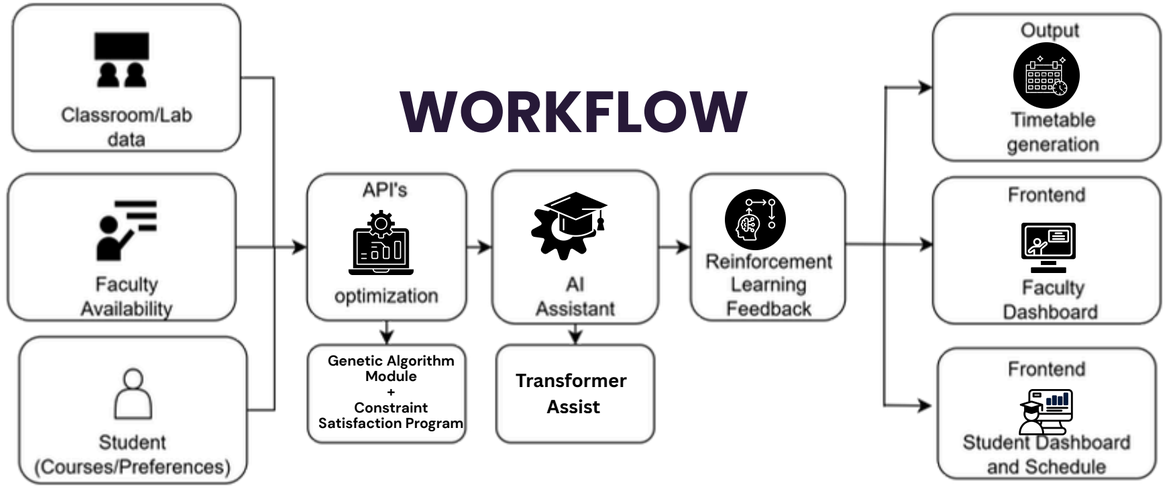


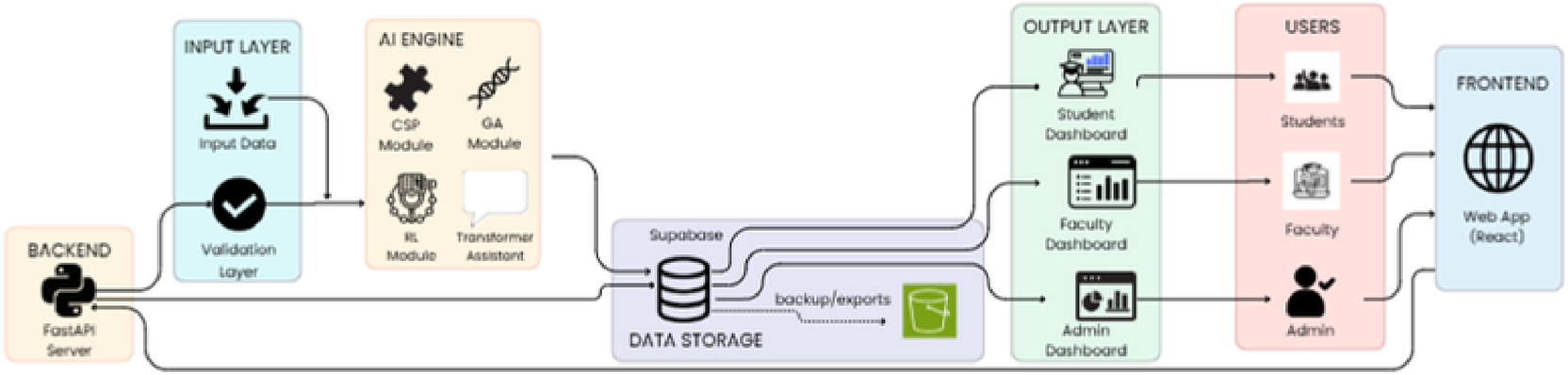
Fig 1.1 - Student dashboard

Fig 1.2 - Admin dashboard

WORKFLOW

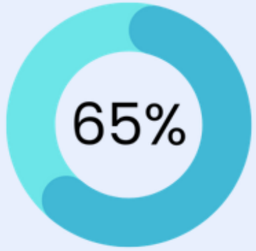


ARCHITECTURE DIAGRAM



PROGRESS

The training of the AI model is in progress with the frontend and authentication being completed at TRL 4



PROTOTYPE LINK

TECH STACK

- PYTHON (FastAPI)
- Algorithms: Steady State Genetic Algorithm
- Constraint satisfaction Problem(CSP)
- Frontend (React)
- Supabase (Database)
- Cloud Deployment (AWS-S3)
- Reinforcement learning module

NEXT STEP FEATURE

Smart Attendance → Auto-tracks via timetable, alerts near threshold while predicting shortages (minimum attendance)

NOTE

AI Engine = Genetic Algorithm + Constraint Satisfaction → generates clash-free timetables with ≥95% accuracy

FEASIBILITY AND VIABILITY

FEASIBILITY

Technical	AI Optimization Time: <2 min per schedule with 99.8% Automated Attendance Accuracy
Operational	Timetable Revision Frequency: 1–2 per semester
Economic	MVP Cost : Rs. 3-6 L ROI : Rs. 2-3 L
Legal Policy	NEP 2020 Alignment and ABC Compliance with no barriers
Scalability	SaaS across states Platform Users: 1,500+ concurrently

CHALLENGES	STRATEGIES
Sudden policy or curriculum revisions disrupt stable schedules..	The modular constraint framework adapts instantly to policy changes.
Late or incomplete data enrollment disrupts timetable generation	Provisional timetables update dynamically as new data arrives.
Faculty workload imbalance creates dissatisfaction and inefficiency	Weighted AI constraints balance faculty workload equitably
Existing ERP systems resist integration with external solutions.	Plug-and-play APIs with shadow mode ease ERP adoption
Solver performance may drop under peak admission load	Cloud parallelization and load balancing sustain performance at scale.

Business Model

Annual Institutional Subscription — Fixed yearly fee per institution (₹5–10 lakhs depending on size) along with partnered revenue from tailored deployments and licensing through ERP integrations.

Schedulo

BENEFITS AND IMPACT

Real Pollution Source Identification

Identifies real pollution sources like stubble, traffic, and industry.

REAL-TIME POLICY IMPACT TRACKING

≥95% clash-free timetables

↓ 25% Faculty Overload

AI-BASED AQI FORECASTING

Balanced workload and reduced admin tasks, more time for mentoring & research

ECO -ACTION REWARD SYSTEM

120+ admin hours saved per semester (per university).

SMART ROUTE OPTIMIZATION

Optimized classrooms & labs, smooth operations.

HIDDEN FINANCIAL GAINS

₹2–3L/yr saved in hidden admin & operational costs.

PERSONALIZED HEALTH & SAFETY ALERTS

Supports NEP 2020 compliance, increased 15% resource use & lower carbon footprint.

FOR SOCIETY

Future-ready, skilled, globally competitive workforce

SDGs

RESEARCH AND REFERENCES

1. DYNAMIC TIMETABLE GENERATOR USING GENETIC ALGORITHM
https://www.ssgmce.ac.in/uploads/UG_Projects/cse/202425/Proj.Gr.-09-Project_Report_2024-25.pdf

2. NEP 2020: A critical examination
https://journals.sagepub.com/doi/abs/10.1177/0049085720958804

3. Planning the Activity Schedule Process Systems by Ai-Based Expert System
https://link.springer.com/chapter/10.1007/978-1-349-09912-2_21

	Schedulo	Tiza	School Timetable.in
Automatic scheduling	✓	✗	✗
Constraint management	✓	✗	✗
Teacher and room management	✓	✗	✗
Drag-and-drop interface	✓	✗	✗

4. Optimizing satisfaction in a multi-courses allocation problem combined with a timetablingproblem
https://link.springer.com/article/10.1007/s00500-016-2375-8

5. Constructing university timetable using constraint satisfaction programming approach
https://ieeexplore.ieee.org/abstract/document/1631445